

Database Management Systems: Assignment-8

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Q1. The dataset consists of four tables: author, book, adaptation, and book_review.

Author (id, name, birth_year, death_year)

Book (id, author_id, title, publish_year, publishing house)

Adaptation (book_id, type, title, release_year, rating)

Book_review (book_id, review, author)

Q	 id int(11)	 name varchar(100)	 birth_year int(11)	 death_year int(11)
>	1	J.K. Rowling	1965	(NULL)
>	2	George Orwell	1903	1950
>	3	Haruki Murakami	1949	(NULL)
>	4	New Author	1980	(NULL)

Q	 id int(11)	 author_id int(11)	 title varchar(200)	 publish_year int(11)	 publishing_house varchar(100)
>	1	1	Harry Potter and the Goblet of Fire	2000	Bloomsbury
>	2	1	Harry Potter and the Deathly Hallows	2007	Bloomsbury
>	3	2	1984	1949	Secker & Warburg
>	4	3	Kafka on the Shore	2002	Shinchosha
>	5	(NULL)	Anonymous Tales	2020	IndiePress

Q	 book_id int(11)	 type varchar(50)	 title varchar(200)	 release_year int(11)	 rating decimal(3,1)
>	1	Movie	Harry Potter and the Goblet of Fire (Film)	2005	7.7
>	2	Movie	Harry Potter and the Deathly Hallows: Part 1	2010	8.1
>	3	Movie	1984 (Film)	1984	7.2

Q	 book_id int(11)	 review text	 author varchar(100)
>	1	A magical continuation of the series.	John Smith
>	3	A chilling dystopian masterpiece.	Alice Brown
>	4	Surreal and deep.	Tom Hanks

- a) Show the name of each author together with the title of the book they wrote and the year in which that book was published.

Query:

```
SELECT a.name AS author_name, b.title AS book_title, b.publish_year FROM Author
a JOIN Book b ON a.id = b.author_id
```

Output:

	author_name varchar	book_title varchar	publish_year int
>	J.K. Rowling	Harry Potter and the Goblet of Fire	2000
>	J.K. Rowling	Harry Potter and the Deathly Hallows	2007
>	George Orwell	1984	1949
>	Haruki Murakami	Kafka on the Shore	2002

- b) Show the name of each author together with the title of the book they wrote and the year in which that book was published. Show only books published after 2005.

Query:

```
SELECT a.name AS author_name, b.title AS book_title, b.publish_year FROM Author
a JOIN Book b ON a.id = b.author_id WHERE b.publish_year > 2005
```

Output:

	author_name varchar	book_title varchar	publish_year int
>	J.K. Rowling	Harry Potter and the Deathly Hallows	2007

- c) For each book, show its title, adaptation title, adaptation year, and publication year.

Query:

```
SELECT b.title AS book_title, ad.title AS adaptation_title, ad.release_year AS
adaptation_year, b.publish_year FROM Book b JOIN Adaptation ad ON b.id = ad.book_id
```

Output:

	book_title varchar	adaptation_title varchar	adaptation_year int	publish_year int
>	Harry Potter and the Goblet of Fire	Harry Potter and the Goblet of Fire (Film)	2005	2000
>	Harry Potter and the Deathly Hallows	Harry Potter and the Deathly Hallows: Part 1	2010	2007
>	1984	1984 (Film)	1984	1949

- d) Show the title of each book together with the title of its adaptation and the date of the release. Show all books, regardless of whether they had adaptations.

Query:

```
SELECT b.title AS book_title, ad.title AS adaptation_title, ad.release_year FROM Book
b LEFT JOIN Adaptation ad ON b.id = ad.book_id;
```

Output:

Q	book_title varchar	adaptation_title varchar	release_year int
>	Harry Potter and the Goblet of Fire	Harry Potter and the Goblet of Fire (Film)	2005
>	Harry Potter and the Deathly Hallows	Harry Potter and the Deathly Hallows: Part 1	2010
>	1984	1984 (Film)	1984
>	Kafka on the Shore	(NULL)	(NULL)
>	Anonymous Tales	(NULL)	(NULL)

- e) Show all books with their movie adaptations. Select each book's title, the name of its publishing house, the title of its adaptation, and the type of the adaptation. Keep the books with no adaptations in the result.

Query:

```
SELECT b.title AS book_title, b.publishing_house, ad.title AS adaptation_title, ad.type
AS adaptation_type FROM Book b LEFT JOIN Adaptation ad ON b.id = ad.book_id;
```

Output:

Q	book_title varchar	publishing_house varchar	adaptation_title varchar	adaptation_type varchar
>	Harry Potter and the Goblet of Fire	Bloomsbury	Harry Potter and the Goblet of Fire (Film)	Movie
>	Harry Potter and the Deathly Hallows	Bloomsbury	Harry Potter and the Deathly Hallows: Part 1	Movie
>	1984	Secker & Warburg	1984 (Film)	Movie
>	Kafka on the Shore	Shinchosha	(NULL)	(NULL)
>	Anonymous Tales	IndiePress	(NULL)	(NULL)

- f) Join the book_review and book tables using a RIGHT JOIN. Show the title of the book, the corresponding review, and the name of the review's author. Consider all books, even those that weren't reviewed.

Query:

```
SELECT b.title AS book_title, br.review, br.author AS review_author FROM
Book_review br RIGHT JOIN Book b ON br.book_id = b.id;
```

Output:

Q	book_title varchar	review text	review_author varchar
>	Harry Potter and the Goblet of Fire	A magical continuation of the series.	John Smith
>	1984	A chilling dystopian masterpiece.	Alice Brown
>	Kafka on the Shore	Surreal and deep.	Tom Hanks
>	Harry Potter and the Deathly Hallows	(NULL)	(NULL)
>	Anonymous Tales	(NULL)	(NULL)

- g) Display the title of each book along with the name of its author. Show all books, even those without an author. Show all authors, even those who haven't published a book yet. Use a FULL JOIN.

Query:

```
SELECT b.title AS book_title, a.name AS author_name FROM Book b LEFT JOIN  
Author a ON b.author_id = a.id UNION SELECT b.title AS book_title, a.name AS author_name  
FROM Book b LEFT JOIN Author a ON b.author_id = a.id;
```

Output:

Q	book_title varchar	author_name varchar
>	Harry Potter and the Goblet of Fire	J.K. Rowling
>	Harry Potter and the Deathly Hallows	J.K. Rowling
>	1984	George Orwell
>	Kafka on the Shore	Haruki Murakami
>	Anonymous Tales	(NULL)

Q2. The dataset consists of five tables : Department, Product, Nutrition_data,

Producer,Sales_history

Department (id, name)

Product(id,name,department_id,shelf_id,producer_id,price)

Nutrition_data(product_id, calories, fat, carbohydrate, protein)

Producer(id, name)

Sales_history(date, product_id,amount)

	  id int(11)	 name varchar(100)
>	1	Beverages
>	2	Snacks
>	3	Dairy
>	4	Bakery

	  id int(11)	 name varchar(100)	 department_id int(11)	 shelf_id int(11)	 producer_id int(11)	 price decimal(10,2)
>	1	Coke	1	101	2	35.00
>	2	Lays Chips	2	102	2	20.00
>	3	Milk	3	103	3	25.00
>	4	Chocolate Bar	2	104	1	45.00
>	5	Bread	4	105	4	30.00
>	6	Energy Drink	(NULL)	106	(NULL)	60.00

	  product_id int(11)	 calories int(11)	 fat decimal(5,2)	 carbohydrate decimal(5,2)	 protein decimal(5,2)
>	1	140	0.00	39.00	0.00
>	2	160	10.00	15.00	2.00
>	3	120	5.00	12.00	8.00
>	4	220	12.00	25.00	3.00
>	5	100	1.00	20.00	4.00
>	6	80	0.00	18.00	1.00

Q	 id int(11)	 	name varchar(100)	 
>	1		Nestle	
>	2		PepsiCo	
>	3		Amul	
>	4		Britannia	

Q	date date	 	product_id int(11)	 	amount int(11)	 
>	2024-12-01		1		500	
>	2024-12-02		2		700	
>	2024-12-03		3		200	
>	2024-12-04		4		150	
>	2024-12-05		5		300	

- a) List all products that have fewer than 150 calories. For each product, show its name (rename the column product) and the name of the department where it can be found (name the column department).

Query:

```
SELECT p.name AS product, d.name as department FROM Product p JOIN
Nutrition_data n ON p.id=n.product_id JOIN Department d ON p.department_id = d.id
WHERE n.calories < 150;
```

Output:

Q	product varchar	 	department varchar	 
>	Coke		Beverages	
>	Milk		Dairy	
>	Bread		Bakery	

- b) For each product, show its name, price, producer name, and department name. Alias the columns as product_name, product_price, producer_name, and department_name, respectively. Include all the products, even those without a producer or department. Also, include the producers and departments without a product.

Query:

```
SELECT p.name AS product_name, p.price AS producer_price, pr.name as  
producer_name, d.name as department_name FROM Product p LEFT JOIN Producer pr ON  
p.producer_id = pr.id LEFT JOIN Department d ON p.department_id = d.id UNION SELECT  
p.name AS product_name, p.price AS producer_price, pr.name as producer_name, d.name  
as department_name FROM Product p RIGHT JOIN Producer pr ON p.producer_id = pr.id  
RIGHT JOIN Department d ON p.department_id = d.id
```

Output:

Q	product_name varchar	producer_price decimal	producer_name varchar	department_name varchar
>	Coke	35.00	PepsiCo	Beverages
>	Lays Chips	20.00	PepsiCo	Snacks
>	Milk	25.00	Amul	Dairy
>	Chocolate Bar	45.00	Nestle	Snacks
>	Bread	30.00	Britannia	Bakery
>	Energy Drink	60.00	(NULL)	(NULL)

c) For each product, display the:

Name of the company that produced it (name the column producer_name).

Name of the department where the product is located (name it department_name).

Product name (name it product_name).

Total number of carbohydrates in the product.

Your query should still consider products with no information about producer_id or department_id.

Query:

```
SELECT pr.name AS product_name, d.name AS department_name, p.name AS  
product_name, n.carbohydrate FROM Product p LEFT JOIN Producer pr ON p.producer_id =  
pr.id LEFT JOIN Department d ON p.department_id = d.id LEFT JOIN Nutrition_data n ON p.id  
= n.product_id;
```

Output:

Q	product_name varchar	department_name varchar	product_name varchar	carbohydrate decimal
>	PepsiCo	Beverages	Coke	39.00
>	PepsiCo	Snacks	Lays Chips	15.00
>	Amul	Dairy	Milk	12.00
>	Nestle	Snacks	Chocolate Bar	25.00
>	Britannia	Bakery	Bread	20.00
>	(NULL)	(NULL)	Energy Drink	18.00